

Production Order Create/Edit — Detailed Design Document (詳細設計書)

001_DD — implements 001_BD (SCR-001), requirements REQ-010–REQ-019.

Document control (改版履歴)

Field	Value
Document ID	001_DD
Category	UI + API
System name	ProductionManagementAI
Subsystem name	Production orders
Work item	WI-002
Implements	001_BD revision 6
Created by	Claude (for ThanhTN)
Created date	2026-09-18
Last updated by	Claude (for ThanhTN)
Last updated date	2026-09-23

Version	Date	Author	Revision content
1	2026-09-18	Claude (for ThanhTN)	Initial creation
2	2026-09-18	Claude (for ThanhTN)	Added the function-design (001_DD-FN) and screen-processing-design (001_DD-SPD) companions; modules 3–5 and flows P-01–P-05 moved there, with pointers left here
3	2026-09-18	Claude (for ThanhTN)	Aligned with implementation (plan revision 2): <code>AllowedNext()</code> extension method; OpenTelemetry instrumentation set; test-plan IDs filled in
4	2026-09-22	Claude (for ThanhTN)	Completion tracking (WI-004 REQ-033, 001_BD revision 6, 003_DB): module 1 sets <code>CompletedAtUtc</code> on <code>InProgress</code> → <code>Completed</code> ; state-transition side effect, database mapping and test viewpoints updated

Version	Date	Author	Revision content
5	2026-09-23	Claude (for ThanhTN)	Diagrams redrawn for the revised DD template (RFC 0009): the layout sketch and the discard dialog as SVG wireframes under <code>wireframes/</code> ; the order status transitions also drawn as a Mermaid state diagram above the table. No field, rule, API or processing changes
6	2026-09-23	Claude (for ThanhTN)	WI-005: the UI is Japanese. The message catalog lists the Japanese text with an English gloss (WI-005 DEC-003); status options, lock hint, required legend, timestamp format (<code>YYYY/MM/DD HH:mm</code> , WI-005 DEC-006), <code>lang="ja"</code> , wireframes and mockup updated. A Japanese edition of the mockup was added (<code>mockups/001_DD_screen-a-mockup.ja.html</code>). No field, rule, API or processing changes

Overview and reference documents (概要・目次)

Field	Value
File / component name	Frontend <code>ProductionOrderPage.tsx</code> (+ feature components); backend <code>ProductionOrdersController</code> , <code>ProductsController</code> , <code>ProductionOrderService</code> , <code>ProductionOrder</code> entity
Overview	Create/edit form for one production order, with status transitions, field locking, stale-save protection and discard confirmation, backed by a JSON API over the 001_DB schema

Module / method / processing index

No	Name	Overview	Notes
1	<code>ProductionOrder</code> (Domain)	Entity + business invariants: transitions, field lock, quantity/notes rules	Unit-tested without a host (0001_ADR)
2	<code>ProductionOrderStatus</code> (Domain)	Enum + allowed-next-states table	Single source of M-02 / V-06
3	<code>ProductionOrderService</code> (Application)	Create / get / update use cases, validation orchestration, error mapping	Designed in 001_DD-FN
4	<code>IPlantClock</code> / <code>PlantClock</code>	"Today" and current year in the plant timezone	Designed in 001_DD-FN
5	<code>IOrderNumberIssuer</code> / <code>OrderNumberIssuer</code> (Infrastructure)	Per-year sequence upsert	Designed in 001_DD-FN
6	<code>ProductionOrdersController</code> , <code>ProductsController</code> (Api)	HTTP endpoints, auth policy, Problem Details	Contract in 001_DD-API
7	<code>ProductionOrderPage</code> (frontend)	Route component: load, mode, states, title	
8	<code>ProductionOrderForm</code> (frontend)	Fields, client validation, dirty tracking, save	
9	<code>DiscardChangesDialog</code> (frontend)	Confirmation on Cancel with unsaved changes	REQ-019
10	<code>apiClient</code> (frontend)	JSON fetch wrapper, Problem Details parsing, 401 handling	Shared

No	Name	Overview	Notes
11	Processing flows P-01–P-05	Load, create, update, cancel, error handling	Designed in 001_DD-SPD

Reference documents

No	Document	Purpose / use	Notes
1	docs/en/010_basic-design/001/001_BD_製造指示登録・編集.md	Screen items, validation V-01–V-08, events E-01–E-09, flows	Item numbers (n) below are 001_BD §3 numbers
2	docs/en/database/001/001_DB_製造指示登録・編集.md (001_DB)	Tables, constraints, counter upsert, xmin , privileges	
3	docs/en/020_detailed-design/001/001_DD-API_製造指示登録・編集.md (001_DD-API)	Endpoint request/response/error catalog	Companion
3a	docs/en/020_detailed-design/001/001_DD-FN_製造指示登録・編集.md (001_DD-FN)	Shared service/clock/issuer method design	Companion
3b	docs/en/020_detailed-design/001/001_DD-SPD_製造指示登録・編集.md (001_DD-SPD)	Per-component processing flows	Companion
4	docs/en/architecture/0001/0001_ADR_backend-layered-structure.md	Layer placement	
5	docs/en/architecture/0002/0002_ADR_auth-rbac-foundation.md	Cookie auth, fallback policy, CSRF (DEC-020)	
6	work-items/WI-002/brief.md , decisions.md	REQs, DEC-001–DEC-024	

Referenced by

No	Document	Purpose / use	Notes
1	001_DD-API, 001_DD-FN, 001_DD-SPD	Companions; each lists 001_DD as its parent	
2	work-items/WI-002/evidence.md	Traceability	

Component / file organization

X-1. Path structure

No	Path / namespace	Purpose	Notes
1	src/backend/ProductionManagementAI.Domain/ProductionOrders/	ProductionOrder , ProductionOrderStatus , Product , DomainRuleViolation	No external dependencies
2	src/backend/ProductionManagementAI.Application/ProductionOrders/	ProductionOrderService , request/response DTOs, IPlantClock , IOrderNumberIssuer , IProductionOrderRepository , ProductionOrderTelemetry	
3	src/backend/ProductionManagementAI.Infrastructure/ProductionOrders/	EF configuration, repository, OrderNumberIssuer , PlantClock ; new migration AddProductionOrders	
4	src/backend/ProductionManagementAI.Api/Controllers/	ProductionOrdersController , ProductsController	
5	src/frontend/src/features/production-orders/	ProductionOrderPage.tsx , ProductionOrderForm.tsx , DiscardChangesDialog.tsx , validation.ts , messages.ts , api.ts , types.ts	
6	src/frontend/src/lib/apiClient.ts	Shared fetch wrapper	New
7	tests/backend/ProductionManagementAI.Application.Tests/ProductionOrders/	Domain + service unit tests	
8	tests/integration/ProductionManagementAI.Integration.Tests/ProductionOrders/	Endpoint tests (Testcontainers Postgres)	
9	src/frontend/tests/unit/production-orders/	Component tests (Vitest + RTL)	

X-2. Shared/common components used

No	Name	Purpose	Notes
1	useAuth / ProtectedRoute (WI-001)	Signed-in user and roles; redirect to /login	Existing
2	apiClient	getJSON / sendJson : same-origin credentials, Content-Type: application/json , parses RFC 9457 bodies into ApiProblem , and on 401 clears the auth state and navigates to /login	New; the WI-001 auth calls keep their current code
3	ASP.NET Core AddProblemDetails() + [ApiController]	Problem Details for all API errors	New registration in Program.cs

X-3. Feature-level components used

No	Name	Purpose	Notes
1	ProductionOrderForm	Items 6–16	Stateless about routing; receives initial values + callbacks

No	Name	Purpose	Notes
2	DiscardChangesDialog	Items 17–19	Native <code><dialog></code> with <code>showModal()</code> (built-in focus trap, Escape)
3	FieldError	Inline error text linked by <code>aria-describedby</code>	
4	MessageBanner	Item 5, <code>role="status"</code> (success) / <code>role="alert"</code> (error)	

X-4. External APIs used — see "APIs used" below.

X-5. Responsive composition — one component tree with Tailwind responsive classes: the `sm:` breakpoint (640px) switches quantity and due date to two columns, and buttons to inline right-aligned. Below 640px everything is one column, and the buttons are full width with Save above Cancel (DOM order Cancel, Save; `flex-col-reverse` on mobile keeps the tab order logical).

Companion design documents

No	Document	Type	Covers
1	001_DD-API_製造指示登録・編集.md	api-design	<code>GET /api/products</code> , <code>POST /api/production-orders</code> , <code>GET / PUT /api/production-orders/{id}</code>
2	001_DD-FN_製造指示登録・編集.md	function-design	<code>ProductionOrderService</code> (list/get/create/update), <code>ProductionOrderMapper</code> , <code>IPlantClock</code> , <code>IOrderNumberIssuer</code>
3	001_DD-SPD_製造指示登録・編集.md	screen-processing-design	P-01–P-05 per component: page, form, save-response handling, discard dialog, controller boundary

Task / design index

No	Category	File-level task	Function/process-level task	Item-level task	Confirmed	Issue	Reviewer	Reworked	Date	Notes
1	basic-design	001_BD	SCR-001 all sections	—	yes	—	ThanhTN	yes	2026-09-18	revision 4 (V-02 upper bound)
2	detailed-design	001_DD	Modules 1–10, P-01–P-05	Items 4–19	yes	—	ThanhTN	yes	2026-09-18	this document; approved 2026-09-18
3	api-design	001_DD-API	4 endpoints	request/response fields	yes	—	ThanhTN	no	2026-09-18	approved 2026-09-18
5	function-design	001_DD-FN	7 methods	—	yes	—	ThanhTN	no	2026-09-18	approved 2026-09-18
6	screen-processing-design	001_DD-SPD	5 processing blocks	—	yes	—	ThanhTN	no	2026-09-18	approved 2026-09-18
4	detailed-design	001_DB	Tables, counter, <code>xmin</code>	—	yes	—	ThanhTN	no	2026-09-18	reviewed by user

Module design

1. ProductionOrder (Domain entity)

Field	Value
Description	Aggregate for one production order. Owns every invariant that doesn't need I/O: status transitions, the product/quantity lock, and quantity/notes value rules
Return type	— (entity)
Created by / date	Claude / 2026-09-18
Last modified by / date	—

Preconditions: none.

Rule type: business invariant.

Rule references

No	Rule / validator	Location	Notes
1	ProductionOrderStatusExtensions.AllowedNext	Domain	V-06

Condition / data references

No	Key	Source	Notes
1	Saved Status , ProductId , Quantity , DueDate	Entity state loaded by the service	V-04 "changed", V-07

Check parameters

No	Name	Type	Target field
1	quantity	int	(9)
2	notes	string?	(11)
3	newStatus	ProductionOrderStatus	(7)
4	productId , quantity vs. saved	Guid , int	(8)(9)

Arguments (factory and mutator)

No	Type	Name	Description
1	Guid	productId	Create / Update
2	int	quantity	1-999,999,999 (V-02, DEC-024)
3	DateOnly	dueDate	Date checks against "today" are done in the service (they need IPlantClock)
4	string?	notes	Trimmed; empty → null ; ≤ 500 Unicode code points (V-05)
5	ProductionOrderStatus	status	Update only
6	short , int	orderYear , orderSeq	Create only, from IOrderNumberIssuer

Dependencies: none.

Processing overview: `ProductionOrder.Create(...)` returns a new `Draft` order. `Update(...)` first checks the lock and the transition, and throws `DomainRuleViolation` with a code (`MSG-E007` , `MSG-E008`) before changing any state. Value-rule failures (quantity, notes) are reported by the static `Validate...` helpers, which return error lists rather than throwing. The service calls them first so all field errors can be returned together.

Processing flow (Update)

Step	Description	Calls
1	If <code>Status != Draft</code> and (<code>productId != ProductId</code> or <code>quantity != Quantity</code>) → throw <code>DomainRuleViolation("MSG-E008")</code> ; nothing changed (DEC-007)	—
2	If <code>status != Status</code> and <code>status ∉ AllowedNext(Status)</code> → throw <code>DomainRuleViolation("MSG-E007")</code>	<code>ProductionOrderStatusExtensions.AllowedNext</code>
3	Assign <code>ProductId</code> , <code>Quantity</code> (only possible while <code>Draft</code>), <code>DueDate</code> , <code>Notes</code> , <code>Status</code>	—
4	Set <code>UpdatedAtUtc</code> = the given UTC time	—
5	If the status moved from <code>InProgress</code> to <code>Completed</code> in this call, set <code>CompletedAtUtc</code> = the same UTC time. Never set otherwise, never cleared; there is no public setter or request field for it (WI-004 REQ-033, 003_DB ck_production_orders_completed_at_matches_status)	—

Return value: none — state change or exception.

2. ProductionOrderStatus (Domain enum)

Field	Value
Description	Draft , InProgress , Completed , Cancelled , plus the AllowedNext() extension method in ProductionOrderStatusExtensions (C# enums can't hold methods), whose table is: Draft → {InProgress, Cancelled}; InProgress → {Completed, Cancelled}; Completed, Cancelled → {}
Return type	IReadOnlySet<ProductionOrderStatus>

Not applicable — a plain lookup table, not a rule/validator with its own fields. Persisted as a string via EF value conversion (DEC-015). The API returns `allowedNextStatuses` so the frontend never duplicates the table (M-02).

3–5. ProductionOrderService, IPlantClock, IOrderNumberIssuer (shared)

These modules will be reused by Screen B, so per `ai/templates/detailed-design.md` they are designed in the function-design companion, **001_DD-FN** (`001_DD-FN_製造指示登録・編集.md`). That document covers the methods, arguments, the fixed check order, telemetry points and value mapping. It is not repeated here.

6. Controllers (Api)

Field	Value
Description	ProductionOrdersController (<code>api/production-orders</code>) and ProductsController (<code>api/products</code>), both <code>[Authorize(Policy = "ProductionOrderEditor")]</code> , where the policy is <code>RequireRole("Admin", "Operator")</code> (DEC-001). Map <code>Result</code> to 200/201/400/404/409/422 Problem Details. <code>[Consumes("application/json")]</code> on POST/PUT (DEC-020)
Return type	ActionResult<T>

Not applicable — plain handler. Route constraint `{id:guid}` , so a non-GUID id returns 404 without reaching the service.

7. ProductionOrderPage (frontend)

Field	Value
Description	Route component for <code>/production-orders/new</code> and <code>/production-orders/:id</code> . Decides the mode, runs P-01, holds page state (<code>loading</code> <code>ready</code> <code>notFound</code> <code>forbidden</code> <code>loadError</code>), sets <code>document.title</code> (0-2-1), shows the banner, and renders <code>ProductionOrderForm</code>
Return type	JSX

Not applicable — plain component. Role check (UX only; the server is authoritative): if `user.roles` includes neither `Admin` nor `Operator`, render the forbidden panel (MSG-E012) without calling the API.

8. `ProductionOrderForm` (frontend)

Field	Value
Description	Controlled form for items 6–16. Keeps <code>initialValues</code> (a snapshot from load or save) and <code>values</code> ; <code>isDirty = !equal(values, initialValues)</code> (with notes compared after trimming). Runs field validators on blur (E-02) and all of them on Save (E-04)
Return type	JSX

Rule type: format check (client-side, advisory; mirrors V-01–V-05 in `validation.ts`, same message IDs).

Check parameters

No	Name	Type	Target field
1	<code>productId</code>	string	(8) non-empty
2	<code>quantity</code>	string	(9) <code>^[0-9]{1,9}\$</code> and ≥ 1
3	<code>dueDate</code>	<code>YYYY-MM-DD</code>	(10) non-empty; $\geq$ browser-local today when creating or changed (DEC-021)
4	<code>notes</code>	string	(11) <code>[...notes].length ≤ 500</code>

Not applicable for V-06/V-07/V-08: the UI prevents V-06 and V-07 (restricted options, read-only fields), and V-08 is server-only.

9. `DiscardChangesDialog` (frontend)

Field	Value
Description	Items 17–19. Native <code><dialog></code> opened with <code>showModal()</code> ; initial focus on 編集を続ける (Keep editing) (<code>autofocus</code>); Escape/ <code>cancel</code> event = Keep editing; on close, focus returns to the 「キャンセル」 (Cancel) button
Return type	JSX

Not applicable — plain component.

10. apiClient (frontend)

Field	Value
Description	<code>getJSON<T>(url)</code> , <code>sendJson<T>(method, url, body)</code> . Always sends <code>credentials: 'same-origin'</code> , plus <code>Content-Type: application/json</code> on bodies. Non-2xx → throws <code>ApiError { status, problem?: ProblemDetails }</code> . 401 → calls the auth context's <code>onUnauthorized</code> (clears the user; <code>ProtectedRoute</code> then redirects to <code>/login</code>)
Return type	<code>Promise<T></code>

Not applicable — plain module.

Screen layout and mockup

Refines the 001_BD §1 wireframe. Changes from BD: the timestamps (13)(14) moved to a footer line inside the card; the locked-field hint text was added; the discard dialog (17–19) is drawn.

3 ホーム / 製造指示一覧 / PO-2026-00042

4 製造指示 PO-2026-00042

5 ✓ 製造指示 PO-2026-00042 を保存しました。

6 指示番号	PO-2026-00042		
7 ステータス	進行中		
8 製品 *	P-1004 — ドライブシャフト read-only		
下書き以外の製造指示では変更できません。			
9 数量 *	250	10 納期 *	2026/10/01
11 備考	2号ラインへ納品。		
			12 9/500
13 14 登録日時 2026/09/18 10:02 · 最終更新 2026/09/18 14:31			

15 キャンセル

16 保存

Changes from 001_BD: timestamps moved to a footer inside the card; locked-field hint added. The discard dialog is drawn separately.

Discard dialog (17–19), centered over a dimmed page:

Page dimmed behind the dialog

17 変更を破棄しますか？

この製造指示の変更は保存されていません。

19

編集を続ける

18

破棄

Focus starts on 編集を続ける (Keep editing); Escape closes the dialog and returns focus to キャンセル (Cancel).

Mockup artifact: <https://claude.ai/artifact/FEo1RG27UjZ6vxFCUjxoHq> (private; the owner shares it from the page's Share menu). Source: [mockups/001_DD_screen-a-mockup.html](#) . It has 11 static states: create empty, product list open (30 items), validation errors, created → edit (Draft), in progress (locked), completed (terminal, overdue), stale save (409), discard dialog, not found, SP create, SP locked.

Japanese edition, with the page captions in Japanese too, linked from the Japanese PDF (WI-005): <https://claude.ai/artifact/QY4W3yuMnnBXxBkN2XioiD> (private). Source: [mockups/001_DD_screen-a-mockup.ja.html](#) .

Region	Contains (field/control)	Notes
Header	(1)(2)	Shared app header (new small <code>AppHeader</code> component, also used on the home page)
Page head	(3)(4)(5)	Breadcrumb collapses to 「< 製造指示一覧」 (< Production orders) below 640px
Form card	(6)–(14)	<code>max-w-3xl</code> centered, <code>rounded-lg border border-gray-200 p-6</code> (same vocabulary as <code>LoginPage</code>)
Actions	(15)(16)	Right-aligned ≥ 640px; full width, stacked, 「保存」 (Save) first < 640px
Dialog	(17)–(19)	<code><dialog></code> with <code>::backdrop</code> dimming

Visual vocabulary follows the existing `LoginPage.tsx` : `gray-900` primary button, `border-gray-300` inputs, `text-red-600` errors. Additions: `bg-gray-50` read-only fields with a lock hint, a green success banner (`bg-green-50 text-green-800`) and a red error banner (`bg-red-50 text-red-800`). Focus rings are `focus-visible:ring-2 ring-gray-900 ring-offset-2` , and all text/background pairs are ≥ 4.5:1 contrast.

Screen item definition

Field	Type	Required	Validation rule	Source (BD ref)
(6) <code>orderNumber</code>	string, response only	—	—	§3 item 6
(7) <code>status</code>	enum select; label in create mode	edit: yes	Options = current + <code>allowedNextStatuses</code> (from API); select disabled when that list is empty	§3 item 7, M-01, M-02, V-06
(8) <code>productId</code>	GUID string, <code><select></code> of 30 products ordered by SKU, label <code>{sku} – {name}</code> (e.g. P-1004 – ドライブシャフト), first option 「製品を選択」 (Select a product)	yes	Non-empty (MSG-E001); server: exists (MSG-E002). <code>disabled</code> - looking but focusable read-only when not Draft (<code>aria-readonly</code> , <code>aria-describedby</code> → lock hint)	§3 item 8, V-01, V-07, DEC-022
(9) <code>quantity</code>	<code><input type="text" inputmode="numeric" maxlength="9"></code>	yes	<code>^[0-9]+\$</code> , 1–999,999,999 (MSG-E003 / MSG-E010); read-only when not Draft	§3 item 9, V-02, DEC-024
(10) <code>dueDate</code>	<code><input type="date"></code>	yes	Non-empty (MSG-E004); ≥ today on create or when changed (MSG-E005). Client uses the browser-local date; the server uses the plant date (DEC-021)	§3 item 10, V-03, V-04
(11) <code>notes</code>	<code><textarea rows="4"></code>	no	≤ 500 code points (MSG-E006); no <code>maxlength</code> attribute (it counts UTF-16 units), so the counter turns red and blocks Save instead	§3 item 11, V-05
(12) <code>counter</code>	text	—	<code>{n}/500</code> , red at > 500, <code>aria-live="polite"</code> announced only at ≥ 450	§3 item 12
(13)(14) <code>createdAt</code> , <code>updatedAt</code>	ISO UTC → <code>formatTimestamp</code> (<code>src/lib/format.ts</code> : <code>Intl.DateTimeFormat('ja-JP', ...)</code> , 24-hour) → 「登録日時 2026/09/18 10:02」 / 「最終更新 ...」 , browser timezone	—	—	§3 items 13–14, M-04
(hidden) <code>version</code>	uint	update: yes	Echoed back on PUT	V-08

Message catalog (`messages.ts` ; the server returns the same IDs as `code`). The other UI strings live in the same module's `labels` object (WI-005 DEC-007):

ID	Text (Japanese, as implemented)	English gloss
MSG-E001	製品を選択してください。	Select a product.
MSG-E002	選択した製品は存在しません。	The selected product no longer exists.
MSG-E003	1以上の整数を入力してください。	Enter a whole number of 1 or more.

ID	Text (Japanese, as implemented)	English gloss
MSG-E004	納期を入力してください。	Enter a due date.
MSG-E005	納期に過去の日付は指定できません。	Due date can't be in the past.
MSG-E006	備考は500文字以内で入力してください。	Notes can't exceed 500 characters.
MSG-E007	このステータス変更は許可されていません。	This status change isn't allowed.
MSG-E008	下書き以外の製造指示では、製品と数量を変更できません。	Product and quantity can't be changed after the order leaves Draft.
MSG-E009	この製造指示は他のユーザーによって変更されました。再読み込みして最新の内容を確認してください。	This order was changed by someone else. Reload to see the latest version.
MSG-E010	数量は999,999,999以下で入力してください。	Quantity can't exceed 999,999,999.
MSG-E011	この製造指示は存在しません。	This production order doesn't exist.
MSG-E012	製造指示を管理する権限がありません。	You don't have permission to manage production orders.
MSG-E013	問題が発生しました。もう一度お試しください。	Something went wrong. Try again.
MSG-E014	利用可能な製品がありません。	No products available.
MSG-I001	製造指示 {orderNumber} を登録しました。	Production order {orderNumber} created.
MSG-I002	製造指示 {orderNumber} を保存しました。	Production order {orderNumber} saved.

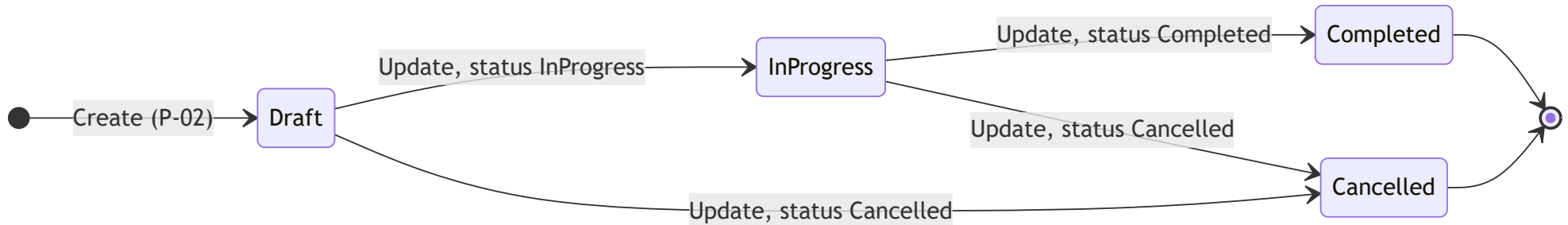
Loading / empty / error / success states

State	Trigger	UI behavior	Data shown
Loading	P-01 in flight	Header, heading placeholder and a skeleton card (<code>aria-busy="true"</code>); no form controls	—
Ready — create	Products loaded	Empty form, status label 「下書き」 (Draft), order number 「保存時に採番」 (Assigned on save)	Product options
Ready — edit (Draft)	Order + products loaded, status Draft	All inputs editable; status options 「下書き」 / 「進行中」 / 「取消」 (Draft/In progress/Cancelled)	Saved values
Ready — edit (locked)	Status InProgress/Completed/Cancelled	Product and quantity read-only with the lock hint 「下書き以外の製造指示では変更できません。」 (Can't be changed once the order leaves Draft); for Completed/Cancelled the status select is disabled too; due date and notes stay editable (DEC-004)	Saved values
Empty products	<code>GET /api/products</code> returns <code>[]</code>	Product select shows MSG-E014 and is disabled; Save shows MSG-E001	—

State	Trigger	UI behavior	Data shown
Saving	Save pressed, request in flight	Save disabled with label 「保存中...」 (Saving...); inputs stay as they are; Cancel still enabled	Entered values
Validation error	Client or 400 response	Inline errors (<code>FieldError</code> , <code>aria-invalid</code>), focus to the first invalid field	Entered values kept
Rule / conflict error	422 (MSG-E007/E008) or 409 (MSG-E009)	Error banner (<code>role="alert"</code>); for 409 the banner has a 再読み込み (Reload) button (re-runs P-01 and discards local edits)	Entered values kept
Not found	404 on load	Panel MSG-E011 with a 「製造指示一覧に戻る」 (Back to production orders) link; no form	—
Forbidden	Missing role (client check) or 403	Panel MSG-E012 with a 「製造指示一覧に戻る」 (Back to production orders) link	—
Load error	Network/5xx on load	Panel MSG-E013 with もう一度試す (Try again) (re-runs P-01)	—
Success	201 / 200	Banner MSG-I001 / MSG-I002 (<code>role="status"</code>); form reset to the saved values, so <code>isDirty = false</code>	Saved values

Processing and state transitions

State transitions



Same-status updates and rejected transitions (422 MSG-E007) are in the table below; they don't change the state.

From state	Event	To state	Side effect
(none)	Create (P-02)	Draft	Row inserted; counter incremented; <code>orders.created</code> metric
Draft	Update with status InProgress	InProgress	Product/quantity become locked

From state	Event	To state	Side effect
Draft	Update with status Cancelled	Cancelled	Terminal
InProgress	Update with status Completed	Completed	Terminal; <code>completed_at_utc</code> set to the save's UTC time (WI-004 REQ-033)
InProgress	Update with status Cancelled	Cancelled	Terminal
any	Update with the same status	same	Other editable fields saved
any other pair	Update	unchanged	422 MSG-E007

Processing flows

The step-by-step processing is in the screen-processing companion, **001_DD-SPD** (`001_DD-SPD_製造指示登録・編集.md`), with one block per component:

Flow	Where	Events
P-01 Load	001_DD-SPD §1 <code>ProductionOrderPage</code>	E-01
P-02 Create / P-03 Update (client)	001_DD-SPD §2 <code>ProductionOrderForm</code>	E-02–E-06, E-08
P-02 / P-03 (server)	001_DD-FN §3 <code>CreateAsync</code> , §4 <code>UpdateAsync</code> ; request boundary in 001_DD-SPD §5	—
P-04 Cancel	001_DD-SPD §4 <code>DiscardChangesDialog</code>	E-07, E-09
P-05 Save-response handling	001_DD-SPD §3	E-06

APIs used

Endpoint	Method	Purpose	Design doc
<code>/api/products</code>	GET	Product options (FN-004)	001_DD-API §1
<code>/api/production-orders</code>	POST	Create (FN-001)	001_DD-API §2
<code>/api/production-orders/{id}</code>	GET	Load (FN-003)	001_DD-API §3
<code>/api/production-orders/{id}</code>	PUT	Update (FN-002)	001_DD-API §4

Database and transaction mapping

Operation	Table(s)	Transaction boundary	Concurrency handling
List products	products (id, sku, name)	none; AsNoTracking	not applicable
Load order	production_orders only (no join; the client resolves the product label from the product list)	none; AsNoTracking for GET	not applicable
Create	production_order_number_counters , production_orders	Explicit transaction: counter upsert + insert, then commit	Counter row lock; unique (order_year, order_seq) backstop
Update	production_orders (+ products existence query when productId changes); on InProgress → Completed the same UPDATE also writes completed_at_utc (003_DB)	Single SaveChanges (implicit transaction) — the status and the completion time commit together	xmin token (DEC-014): stale → 409

Queries select only the columns used (projection to DTOs). The runtime login is pmai_app , with the grants in 001_DB (DEC-016).

Exception handling

Failure	Retry policy	Timeout	User-facing error	Logging
Validation (400)	none	—	Inline field errors	Information: ProductionOrderValidationFailed with order id (if any), user id, failed field names (not values)
Business rule (422)	none	—	Banner MSG-E007/E008	Information: ProductionOrderRuleViolated with code, from/to status
Stale save (409)	none (user reloads)	—	Banner MSG-E009 + 「再読み込み」 (Reload)	Information: ProductionOrderConcurrencyConflict
Not found (404)	none	—	Panel MSG-E011	Debug
Counter overflow / DB error / unhandled	none automatic	Npgsql command timeout 30s (default)	Banner or panel MSG-E013	Error with exception (server-side only); response is generic Problem Details with no stack trace or exception text (ai/rules/backend.md)
Transient connection failure	EF Core EnableRetryOnFailure is not enabled: the create uses an explicit transaction, which would need an execution-strategy wrapper; not worth it for the demo	—	MSG-E013; user retries	Error

Failure	Retry policy	Timeout	User-facing error	Logging
Frontend network failure	none automatic; user presses 「保存」 (Save) / 「もう一度試す」 (Try again)	browser default	MSG-E013	<code>console.error</code> only in development

Never logged: notes text, full request/response bodies, cookies.

Observability (OpenTelemetry)

Per `ai/rules/backend.md` and the design-consistency checklist. The backend has no OpenTelemetry setup yet, so plan revision 2 adds `OpenTelemetry.Extensions.Hosting` with ASP.NET Core and Npgsql instrumentation and the OTLP exporter. Npgsql spans cover the database statements; EF Core's own instrumentation package is still prerelease and the backend makes no outgoing HttpClient calls, so neither is added. The exporter is enabled only when `OTEL_EXPORTER_OTLP_ENDPOINT` is set; otherwise telemetry is collected but not exported.

Signal	Name	Attributes / tags	Emitted by
Span (auto)	POST <code>api/production-orders</code> , PUT <code>api/production-orders/{id}</code> , GET ...	<code>http.route</code> , <code>http.response.status_code</code>	ASP.NET Core instrumentation
Span (auto)	DB statements	<code>db.system=postgresql</code> (no parameter values)	Npgsql instrumentation
Span	<code>ProductionOrder.Create</code>	<code>production_order.id</code> , <code>production_order.number</code> , <code>outcome</code>	<code>ProductionOrderService</code> (ActivitySource <code>ProductionManagementAI.ProductionOrders</code>)
Span	<code>ProductionOrder.Update</code>	<code>production_order.id</code> , <code>status.from</code> , <code>status.to</code> , <code>outcome</code>	same
Counter	<code>pmai.production_orders.created</code>	<code>outcome</code> = <code>success</code> <code>validation_failed</code> <code>error</code>	Meter <code>ProductionManagementAI.ProductionOrders</code>
Counter	<code>pmai.production_orders.updated</code>	<code>outcome</code> = <code>success</code> <code>validation_failed</code> <code>rule_violation</code> <code>conflict</code> <code>not_found</code> <code>error</code>	same
Counter	<code>pmai.production_orders.status_transitions</code>	<code>from</code> , <code>to</code>	same, on successful status change
Histogram	<code>http.server.request.duration</code> (auto)	<code>route</code> , <code>status</code>	ASP.NET Core

The user id is not a span attribute (it's personal data). Logs correlate with spans through the trace id.

Configuration

Key	Default	Notes
Plant:TimeZone	Asia/Tokyo	IANA ID; validated at startup (DEC-017)
ConnectionStrings:DefaultConnection	—	Runtime login pmai_app (DEC-016); secret from environment
OTEL_EXPORTER_OTLP_ENDPOINT	unset	Optional

Accessibility (WCAG 2.2 AA)

- Every input has a `<label for>`. Required inputs have `aria-required="true"` and a visible `*` with the legend 「は必須項目です。」 (marks a required field).
- Errors: `aria-invalid="true"` + `aria-describedby` → the `FieldError` id. On a failed save, focus moves to the first invalid field.
- Read-only locked fields: native `readonly` on the quantity input. The product `<select>` can't be `readonly`, so it's rendered as a read-only text input showing the label, with `aria-describedby` → the lock hint. It stays focusable, unlike `disabled`.
- Banners: success `role="status"`, errors `role="alert"`. Panels get focus on render (`tabIndex=-1`).
- Language: `<html lang="ja">`, so assistive technology reads the Japanese labels with a Japanese voice (WI-005).
- Dialog: native `<dialog>` + `showModal()` (focus trap, Escape). It is labelled by its title via `aria-labelledby`.
- Focus-visible rings on every control; target size ≥ 24×24 CSS px (2.5.8); contrast ≥ 4.5:1 for text and ≥ 3:1 for control borders.
- Keyboard: the natural tab order matches the visual order; native date input and select.

Test viewpoints and unresolved decisions

Level: U = unit (xUnit Domain/Application, or Vitest+RTL for the frontend), I = integration (`WebApplicationFactory` + `Testcontainers Postgres`), E = E2E (Playwright, DEC-025). Test-plan IDs refer to `work-items/WI-002/test-plan.md` (TP-002).

Scenario	Precondition	Expected result	Test-plan ID
Create valid order (REQ-010) — I, E	Admin signed in, products seeded	201, <code>status=Draft</code> , <code>orderNumber</code> matches <code>^PO-\d{4}-\d{5}\$</code> , redirect to edit + MSG-I001	TC-001
First order of a year gets 00001 ; next gets 00002 (DEC-012/013) — I	Empty counter; plant clock faked to 2026 then 2027	PO-2026-00001 , PO-2026-00002 , PO-2027-00001	TC-002

Scenario	Precondition	Expected result	Test-plan ID
Concurrent creates get distinct numbers — I	20 parallel POSTs	20 distinct sequential numbers, no gaps	TC-002
Year boundary uses the plant timezone (DEC-017) — U	<code>TimeProvider</code> at 2026-12-31T15:30Z (= 2027-01-01 00:30 JST)	Year 2027; "today" 2027-01-01	TC-003
Edit Draft order fields (REQ-011) — I	Draft order	200, values saved, <code>updatedAt</code> changed	TC-004
Unknown id (REQ-011) — I, U(fe)	—	404 MSG-E011; not-found panel	TC-005
Stale save (DEC-010) — I	Two clients load version v; A saves	B's PUT → 409 MSG-E009; row keeps A's values	TC-006
Unauthenticated (REQ-012) — I, U(fe)	No cookie	401 on all 4 endpoints; UI redirects to <code>/login</code>	TC-007
No role (REQ-012) — I, U(fe)	User without Admin/Operator	403 on all 4 endpoints; forbidden panel without API call	TC-008
Operator allowed (DEC-001) — I	Operator signed in	200/201	TC-009
Quantity rules (REQ-013) — U, I	—	<code>1</code> , <code>250</code> , <code>999999999</code> pass; empty, <code>0</code> , <code>-5</code> , <code>2.5</code> , <code>abc</code> , <code>1000000000</code> → MSG-E003/E010	TC-010
Due date rules (REQ-014, DEC-009) — U, I	Plant today = D	D and D+1 pass; D-1 → MSG-E005 on create; overdue order saved without changing the due date → 200; changing it to D-1 → MSG-E005	TC-011
Product rules (REQ-015) — I	—	Missing → MSG-E001; random GUID → MSG-E002	TC-012
Notes rules (REQ-016) — U, I	—	Empty → <code>null</code> ; 500 chars (incl. emoji counted as 1) pass; 501 → MSG-E006	TC-013
Each allowed transition (REQ-017) — U, I	Order in the source state	200, new status; <code>allowedNextStatuses</code> updated	TC-014
Completion time recorded (WI-004 REQ-033) — U, I	InProgress order; clock pinned	Saving <code>Completed</code> sets <code>completed_at_utc</code> = the save time (= <code>updated_at_utc</code>); every other transition and every same-status save leaves it unchanged (<code>NULL</code> , or the original value on a later edit of a completed order's notes); a rejected save (422, 409) sets nothing; a request body carrying a completion field is ignored	WI-004 TP-004 (ID assigned in 003_DD)
Disallowed transitions (REQ-017) — U, I	e.g. Draft→Completed, Completed→InProgress, Cancelled→Draft	422 MSG-E007, status unchanged	TC-015

Scenario	Precondition	Expected result	Test-plan ID
Status options per state (M-02) — U(fe)	Each status	Select options match; disabled for terminal states	TC-016
Locked-field change rejected whole (REQ-018, DEC-007) — U, I	InProgress order; PUT changes quantity + notes	422 MSG-E008; notes not saved	TC-017
Locked fields read-only in UI (REQ-018) — U(fe)	InProgress order	Product/quantity read-only, focusable, lock hint	TC-018
Cancel without changes (REQ-019) — U(fe), E	Pristine form	Navigates home, no dialog	TC-019
Cancel with changes → Discard / Keep editing / Escape (REQ-019) — U(fe), E	Edited form	Dialog; Discard → home; Keep editing/Escape → values kept, focus on Cancel	TC-020
Non-JSON body (DEC-020) — I	POST <code>text/plain</code> / form-encoded	415, nothing created	TC-021
Problem Details shape — I	Any 4xx/5xx	<code>application/problem+json</code> , <code>type</code> , <code>title</code> , <code>status</code> , <code>code</code> ; no stack trace	TC-022
Runtime login privileges (DEC-016) — I	App connected as <code>pmai_app</code>	All flows work; <code>DELETE FROM production_orders</code> → permission denied	TC-023
Telemetry emitted — I	In-memory exporter	Create/Update spans + counters with the expected tags	TC-024
Accessibility — U(fe) + manual	—	Labels, <code>aria-invalid</code> / <code>describedby</code> , focus on first error, dialog focus; automated axe check (vitest-axe + @axe-core/playwright, DEC-026) has no violations	TC-025

Unresolved decisions: none. DD-level technical decisions DEC-021–DEC-024 are recorded in `work-items/WI-002/decisions.md`.